**PH.45130** **FALL 2026**

# Quantum Gravitation

## Problem Set 15

Yuran Seo

Due 2026-12-13 23:59

Topic

Final Project Presentations

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

## Problems

1. Derive the first displayed relation used in Final Project Presentations. State all assumptions and conventions.
$$m_g^{95} \sim 10^{-32} eV/c^2$$
2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.
$$checksum_{66}^{after} = pass$$
3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.
$$artifact = hash(pdf, env, log, table, data)$$

Show enough intermediate work that another student can reproduce the calculation.